

STATISTICAL ANALYSIS PLAN FOR THE NSRI2020 SURVEY

VERSION 7

26 January 2021

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PART 1: Key principles for the analysis

1. On the Open Science Framework, a folder named “Data Analysis” will be created containing the original data file ($m=0$) and a single R Markdown script containing the syntax for data imputations and the results for all subsequent analyses. The datafile created by the MICE multiple imputation procedure in R, containing the original data ($m=0$) and the multiply imputed files ($m=1$ to 50), will be also stored there (in .sav and .xlsx format) to facilitate usage by persons who prefer working with other software packages than R. These files will be embargoed until July 2021 (or longer if the core research team deems this necessary). This analysis plan will be uploaded to the Open Science Framework *before* Kantar sends us the anonymized data. This will be verifiable by comparing the OSF upload date with Kantar’s send date. Statistician (GV) will publish all his work from start to finish on a public GitHub repository that will be linked to OSF.
2. Imputations will be performed by a single statistician (GV).
3. The main analysis will be performed independently in duplicate by two statisticians (MC and GV), based on the principles laid out in this document. Inconsistencies between these analyses will be discussed and solved, after arbitration by the core research team members, if needed.
4. The analyses will start with basic descriptions, increase in complexity (adjustments, effects in subgroups) by exploring a number of pre-specified 2-way and 3-way interactions between background and explanatory variables. If these models converge and do not yield standard errors > 100 times the corresponding regression coefficients, we will report their results. These pre-specified interactions were chosen with an eye to actionability and policy-making.
5. The raw data that Kantar will send us will contain a variable identifying whether a participant came from a supportive or non-supportive institute and a variable created by subtracting ‘time of email invitation’ from ‘time of receipt of the reply to the survey’. There will be no identifying variables at the institution or department levels. This implies that neither within-institution nor within-departmental clustering of observations will be accounted for. This is a price we pay for high level of privacy of the participants and their institutions. Some estimated variances may be (slightly) underestimated due to this.
6. The decisions on which independent variables will be included in the regression models will be described below. No automated variable selection techniques will be used.
7. The R-CRAN software package (most recent version at the time of data analysis) will be used for all analyses.
8. A separate publication plan will be formulated and regularly updated. In all publications a clear separation will be made between results based on this data-analysis plan and results based on ideas that emerged later and were therefore potentially data-driven.

9. Deviations from the analysis plan as stipulated in this document will be logged by the two analysts. The same applies to decision taken to reach consensus should the analysts reach different results.
 10. Kantar will provide us with their randomization script. This will be scrutinized only if the group sizes for the three randomized groups were large (missingness-by-design).
 11. Before the multiple imputation steps, basic data editing will be performed to correct obvious errors in the data. One important editing step will involve rescaling all seven survey questions that have been framed counter the direction of the scale (for reasons of linguistic variability and avoiding a monotonous list of questions). These items are called 'contraindicated' and marked as such in an Excel file titled "NSRI -Format variables and values + NA questions". This file will be available to the two statisticians performing the analyses independently from each other.
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PART 2: Imputation plan

Questionnaire set

There are four blocks of questions in the set and two variables related to sampling and propensity to participate:

Block	# items	measurement	NA allowed
Background variables	4	mixed	0
Explanatory variables	75 variables (3 random sets of 50, by missingness by design) in 12 scales	7-Likert	29
QRP/RRP outcomes	22	7-Likert	22
RM outcomes	2	Dichotomous	2
Support-non-support	1	Dichotomous	0
Response delay (days)	1	Continuous	0

Each of the three 50-item subsets contains a different set of items (randomly drawn from the 75 items constituting the 12 scales) that may result in missing values.

A. The 75 explanatory variables are divided into 3 partially overlapping sets of questions, such that every participant was randomly assigned to one of three 50-item subsets. Therefore, for every participant only $\frac{2}{3}$ of the explanatory variables are observed. Participants may answer ‘not applicable’ to all or to a subset of explanatory variable items, which is also coded as such in the data set.

B. Participants may answer ‘not applicable’ to all questions in the research practices block (dependent variables).

C. Participants may answer ‘not applicable’ to all questions in the randomized response block (dependent variables).

Skipping questions

Participants are not allowed to skip questions and instead are forced to answer and continue with the next question or to drop out of the survey altogether.

Not allowing participants to skip questions implies that there is no item non-response due to skipping questions. However, it may impact the quality of the collected data.

Not Applicable

For the explanatory variable scales, participants may answer 'not applicable' whenever the item does not apply to them. These are bona fide values and, as such,

1. will not be set to missing
2. will serve as a basis for estimation and imputation for planned missingness.

Consequently, after solving the missingness, an imputed data set may contain more 'not applicables' than found in the set containing the observed data.

For the outcomes, QRPs and RRP and the two the randomized response items (fabrication and falsification), 'not applicables' will be replaced by the value 1 ("Never") after multiple imputations have been completed. This implies that we interpret 'NA' on these items as "behavior has not been performed" lumping the reasons ('situation does not occur in my work' with 'situation might have occurred but I did never perform that behavior over the last 3 years').

Scope of the missingness

We identify the following types of missingness:

- Dropout: an eligible participant that gave informed consent may choose to exit the survey at any point.
- Planned missingness: participants are randomly assigned to one of three 50-item subsets of explanatory items. For every participant only $\frac{2}{3}$ of the explanatory items are observed.

The planned missingness in the explanatory variables is completely at random (MCAR). This implies that the process of the planned missingness would yield an equal probability of being unobserved for every unobserved item in the explanatory variables block. MCAR missingness is ignorable, meaning that the process that generated the incomplete values does not need to be explicitly modeled when analyzing the incomplete data set.

Dropout will not be dealt with. Note, again, that there is no item non-response in this survey.

Solving the missingness

Let us assume the classical research scenario where a researcher operationalizes a problem, formulates a research question, collects data, analyzes the data and finds an answer to the research question. Any occurrence of missing values in the acquired dataset may have serious consequences for the analysis phase, as most analyses require the data to be completely observed.

To allow for the missing data during analysis, one of two things needs to happen:

1. the estimation methods must be adapted to deal with incomplete data (i.e. missing values need to be integrated out of the analysis model)
2. The missingness needs to be dealt with before the analysis takes place.

Imputation

Imputation follows the second path and is aimed at solving the missing data problem before analysis takes place. With imputation, some estimation procedure is used to impute (fill in) each missing datum, resulting in a complete dataset that can be analyzed as if the data were completely observed.

Multiple imputation

When only one value is imputed (single imputation), uncertainty about the imputations is not reflected in the imputed data set and specific methods for variance estimation that take imputed values into account need to be employed. As a more versatile way to solve this, uncertainty about the imputed values can be taken into account by performing multiple imputation (MI). With MI, each missing datum is imputed $m \geq 2$ times, resulting in m completed datasets. At least 2 imputations are warranted to reflect the uncertainty about the imputations, although performing more imputations is often advisable. The m datasets are then analyzed by standard procedures and the analyses are combined into a single inference, averaging the coefficients and applying Rubin's rule to obtain the correct variances (standard errors and 95% confidence limits).

Why imputation

Accommodating for the missingness in each of the analyses - i.e. analyses based on a subset of items and not on the data set as a whole - would be a) computationally challenging, b) practically unfeasible, and c) methodologically limiting in terms of ignorable missingness. For example, the substantive model of interest of any analysis may not contain relevant features with respect to the missingness.

For reasons listed above, we choose to multiply impute the whole data set and base any subsequent analyses on these multiply imputed data sets. This process is:

- fully reproducible (given a random seed)
- transparent
- without subjective modeling choices for the missingness for sub-analyses

Imputation plan

Imputation method

Observations are realizations of the true data generating model. Our aim is to stay as close to the true data generating model as possible. One technique that has been proven to be particularly useful in this respect is predictive mean matching: a donor-based hot deck technique imputation technique that matches unobserved values with likely observed donor candidates based on their covariates.

Imputation model

We set out to have a fully conditionally specified imputation model where every incomplete column in the data is imputed based on all other columns. For this at least 1560 participants without dropout (104 columns times 15 participants) are needed.

Imputation software

We impute the data with the mice package in R which is the de facto standard for iterative multiple imputation.

Number of iterations

We will use 15 iterations to reach convergence as the default. Visual inspection of the algorithmic convergence through trace plots will guide decisions to increase that value.

Number of imputed data sets

The number of imputed data sets will be 50, or at least 100 times the largest value of the Fraction of Missing Information (FMI).

Further Evaluation

There is one design feature that would result in missingness and may influence the resulting analyses: planned missingness.

Planned missingness

Subgroups were created (after answering the questions on background variables, see Q9009 in the survey) by randomly assigning 50 out of 75 questions on the exploratory variable scales to each potential respondent. Although this process is likely to yield subgroups that are comparable on all background characteristics, differences between these groups may exist:

1. if randomization was flawed
2. if dropout was selective (e.g. by an item perceived to be annoying and not present in all groups)
3. by chance

Flawed randomization will be checked by critically assessing the process, i.e. the logic of the randomization script made by Kantar will be scrutinized by the two statisticians independently. In addition, the relative equality of the sizes of the three randomized groups will be checked. Only gross relative discrepancies may impel us to explore potential reasons at the level of (single) questions. We do this in order to exclude the possibility that e.g. a question that annoyed respondents and that was not assigned to all groups led to selective dropout.

Differences due to chance are ignorable.

All regression models (see below) will contain a *base set* of 5 background variables coding for field, rank, gender, non-empirical research and institutional support via 8 or 9 dummy variables in total. Therefore, models will correct for any imbalances in these variables between the three randomized groups.

Sensitivity analyses may be performed should scenarios not covered in this text emerge.

PART 3: Explanatory variable scales

Construction of explanatory scales

Imputation of the missingness-by-design results in 50 complete datasets. For each of these data sets a series of twelve Principal Components Analyses (PCA) will be performed on the items pertaining to each of the twelve explanatory scales to obtain the individual scores on these scales. In each complete data set, for each scale, Cronbach's Alphas will be calculated, as well as the average Cronbach's Alpha over the complete data sets. Regardless of the magnitude of these alphas, the first component will be retained in the ensuing analyses (based on the regression method applied to standardised item scores).

For the construction of each scale within each complete data set, respondents who answered to all questions with 'non applicable' will be subjected to listwise deletion. For respondents who answered all questions with a score of 1 to 7, the estimated first component score will be used. For respondents with between one 'not applicable' and 1 minus the total number of questions answered 'not applicable', the mean of the observed standardised item values will be used. For each scale, we will check if, and by what extent, the component scores differ between respondents with no 'not applicable' and those with at least one 'not applicable'. Should differences of considerable size occur, we will investigate the potential reasons why they occur and attempt to find appropriate solutions.

Descriptive statistics of the explanatory variable scales

1. For each of the 12 explanatory variables scales, we will calculate the means scale scores by disciplinary field, academic rank, gender, institutional support and engagement in empirical research: $12 \times 14 = 168$ means.
 2. After dealing with occurrences of 'not applicable' as described in this section, above, a 12x12 Pearson correlation matrix of mean scores of explanatory variable scales will be produced, averaged across the 50 complete datasets.
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PART 4: Main analysis

A. Overall descriptive statistics of outcomes

1. Relative prevalences for scores (1,2,3,4,5,6,7,NA) for each RRP
2. Relative prevalences for scores (1,2,3,4,5,6,7,NA) for each QRP
3. Mean scores and standard deviations for each RRP
4. Mean scores and standard deviations for each QRP
5. Mean score and standard deviation across all 11 RRP*
6. Mean score and standard deviation across all 11 QRP*
7. Estimated prevalence of any QRP (i.e. at least one out of 11)*
8. Estimated prevalence of F1 or F2 (taking into account randomized response)*
9. Estimated prevalence of fabrication (F1, taking into account randomized response)
10. Estimated prevalence of falsification (F2, taking into account randomized response)

*Note: All of the above overall statistics will also to be calculated separately for disciplinary fields (4 categories), academic ranks (3 categories), gender (2 categories), (non)supporting institution (2 categories) (2 categories), and engaging in empirical research (2 categories). * Regression analyses will be restricted to the 4 outcomes marked with an asterisk.*

B. Descriptive statistics of the four background variables

1. Absolute and relative frequencies of the 4 background factors: disciplinary field (4 categories), academic rank (3 categories), gender (2 categories) and engagement in empirical research (2 categories), (non)supporting institution (2 categories)
2. Field by rank (12 cells)
3. Field by rank by (non)supporting institution (24 cells)
4. Fully stratified (96 cells).

C. Multiple regression analyses for outcomes A5 – A8

The table below specifies 80 regression analyses, 20 for each of 4 dependent variables.

Dependent variables:

1. RRP mean (A5), linear model
2. QRP mean (A6), linear model

3. Frequency of any QRP (A7), binary logistic model
4. Frequency of F1 or F2 (A8), binary logistic model (taking into account randomized response)

Nr.	Independent variable	Adjustment variables	Comment
1	-	base set	estimate effects of base set variables
2	scale 1	base set	estimate effect of explanatory scale 1
..	..	..	..
13	scale 12	base set	estimate effect of explanatory scale 12
14	scale 1 - 12	base set	estimate effect of all explanatory scales simultaneously
15	field * rank	base set	learn if the effect of rank, if any, varies by field
16	competition * field	base set + separate variables that make up the interactions	learn if the effect of competition, if any, varies by field
17	detection * field	base set + separate variables that make up the interactions	learn if the effect of beliefs about detection, if any, varies by field
18	funding pressure * rank	base set + separate variables that make up the interactions	learn if the effect of funding pressure, if any, varies by academic ranks
19	publication pressure * rank	base set + separate variables that make up the interactions	learn if the effect of publication pressure, if any, varies by academic ranks

1. All multiple regression models contain the base set variables using 8 or 9 indicator ('dummy') variables, 3 dummies for field [biomedical = reference], 2 dummies for rank [postdoc etc category = reference], 2 dummies for gender [males = reference; 1 dummy if there are no gender-neutral answers], 1 dummy for (non)supporting institution [No support = reference] and 1 dummy for non-empirical research is included, empirical = reference.

2. All interaction models will also contain the separate variables that make up the interactions (hierarchy principle).

3. Asterisks (*) indicate interaction.

4. To decide if interactions exist and before interpreting any individual regression coefficients belonging to variables that constitute the interactions, the ANOVA tables will be assessed to see if the interactions as a whole statistically significantly improve the fit of the model at the $\alpha = 0.10$ level. Epidemiologists may know this approach as the 'chunk test' approach (see David Kleinbaum).

At a later stage we may explore associations with single survey questions as dependent variables instead of taking sum scores across the 11 RRP (outcome A5) and the 11 QRP (outcome A6).

PART 5: Structural equation model (MIMIC SEM)

-Scientific norm subscription	9 items
-Peer Norms	9 items
-Perceived Work pressure	3 items
-Publication pressure	6 items
-Pressure due to dependence on funding	6 items
-Competitiveness of the field	6 items
-Mentoring (Survival)	6 items
-Mentoring (Responsible)	6 items
-Distributional organizational justice	6 items
-Procedural organizational justice	6 items
-Likelihood detection of QRPs by collaborators	6 items
-Likelihood detection of QRPs by reviewers	6 items
-11 QRPs	
-11 RRP	
-2 Research misconduct items	

The analyses described in this chapter will be performed by two investigators independently from each other.

Using the Lavaan package (Rosseel, 2012) version 0.6-7, we will estimate a series of Structural Equation Models (SEMs) with latent variables representing different notions of how the 11 QRP items, the two misconduct items, and the 11 RPR items are related to each other and to the 12 explanatory variables. This allows us to test various hypotheses on the multivariate relations between the research practice items and explanatory variables in the NSRI and to delve deeper in potential group differences. Groups in the SEMs are based on academic rank (3 levels) and disciplinary field (4 levels). The SEM analyses will consist of three steps and will be estimated using the weighted least squares means and variance adjusted (WLSMV) estimator in Lavaan that appropriately treats the research practices as having been measured on either a seven-point ordinal scale or as a combination of the two dichotomous misconduct items subject to noise (because of the randomized response).

Integrity: one, two, or more factors?

In Step 1, we will fit three latent variable SEMs to study the dimensionality of the 23 research practice items (11 QRPs, 11RRPs, and the combined RM item). Model 1 presents a one-factor model (Figure 1) that assumes a single latent dimension of integrity on which the 11 QRP and 2 research misconduct items (combined into one item with observed values 0, 1, or 2 subjected to randomized response) show negative loadings and the 11 RRP items show positive loadings. If this model fits the data, it will support the notion that all 24 research practices can be scaled in a single dimension. However, given the clear distinction in terms of integrity between QRP/RM items on the one hand and the RRP items on the other hand, we will also fit a competing model (Model 2; Figure 2) with two correlated factors representing the QRP/RM factor and the responsible research factor (RRP), in which we expect a negative correlation between both factors. In both models of Figure 1 and 2, we specify correlated residuals between the following pairs of RRP: open data & open code, avoiding errors & correcting errors, and fair authorship & providing credit. In this way, we accommodate covariation between these item pairs also reflecting specific content (open science, errors, and authorship) alongside the RRP factor (or the overall integrity factor in the one factor model). Given that the models are quite restrictive and power to reject misfit is large in light of the large sample size in the NSRI, we will consider the model to fit the data if it meets the standard criteria of approximate fit of $RMSEA < .08$ and $CFI > .90$. However, if neither of the models meet these standards, we will allow at most five modifications to improve fit based on the five largest test score results (modification indices), using the method developed by Mansolf and colleagues in the context of SEM with multiple imputation (Mansolf, Jorgensen, & Enders, 2020). Model misfit will thus allow us to adapt the model slightly using cross loadings (in model 2) and residual covariances (models 1 and 2). If needed, we will choose between the (adapted) and nested models 1 and 2 using the likelihood ratio test (against $\alpha = .005$) and the AIC comparative fit statistic. If after these adaptations neither model 1 nor model 2 meets the thresholds for good fit based in RMSEA and CFI, we will attempt to find a well-fitting explorative model of either 3 or 4 factors underlying the 24 research practices. So if needed, we will fit an explorative factor model with three or four common factors subject to oblique PROMAX rotation (Hendrickson & White, 1964) leading to a simple structure.

Is the latent model the same across academic ranks and disciplinary fields?

Step 2, we will study group differences in the multivariate relation and mean differences in the 23 practices in the latent factor model established in Step 1 to better understand whether the differences in research practices between academic ranks (3 levels) and disciplinary field (4 levels) are specific (i.e., only related to particular research practices) or general (i.e., related to the general factors). Specifically, we will test for strong factorial invariance of the chosen latent variable model in step 1 (a one, two, or three factor model) of the 24 research practices across academic rank (3 levels) and disciplinary field (4 levels). This is done in a stepwise fashion (configural invariance, metric invariance, strong factorial invariance) using the delta

parametrization for categorical variables in Lavaan. This analysis can determine whether specific research practices should *not* be compared across ranks or disciplinary fields because they exhibit group difference in measurement parameters that highlight group-specific differences that cannot be accounted for by the latent factor(s) (Wichert & Dolan, 2010). This is relevant for when we relate the factor(s) to the explanatory variables, and is required to meaningfully compare groups on the latent factor means. If measurement invariance restrictions fail based on the omnibus likelihood ratio test (against Alpha = .001), we will consider sources of misfit using the score test (against Alpha = .001 for individual parameters) and freely estimate the measurement parameters of research practices showing non-invariance in the analyses in Step 3.

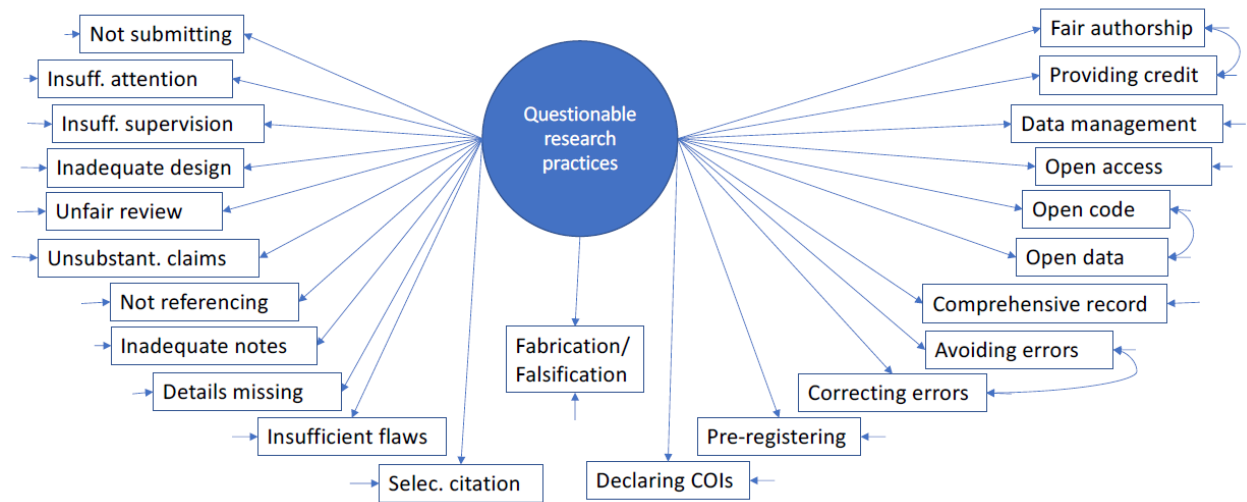
How is/are the integrity factor(s) related to the explanatory variables?

In Step 3, we will fit a multi-group multiple-indicator multiple-cause (MIMIC) models (see Figure 3) with the chosen model from Steps 1 and 2 to study how the explanatory variables relate to the factor or factors and/or to the research practices directly. In the model MIMIC 1, we will regress all (correlated) explanatory variables on the integrity factor(s) and subsequently test for invariance of the regression weights across the groups based on academic rank (3 levels) and disciplinary field (4 levels) based on an omnibus likelihood ratio test. We again use the score tests (modification indices) to assess potential differences between the groups in how the explanatory variables relate to the integrity factor(s). In addition, we will examine using the same score tests whether any of the explanatory variables have any direct effect on the research practices. As a decision rule, the score test should exceed 10.82 (the critical value for the DF = 1 test at Alpha = .001).

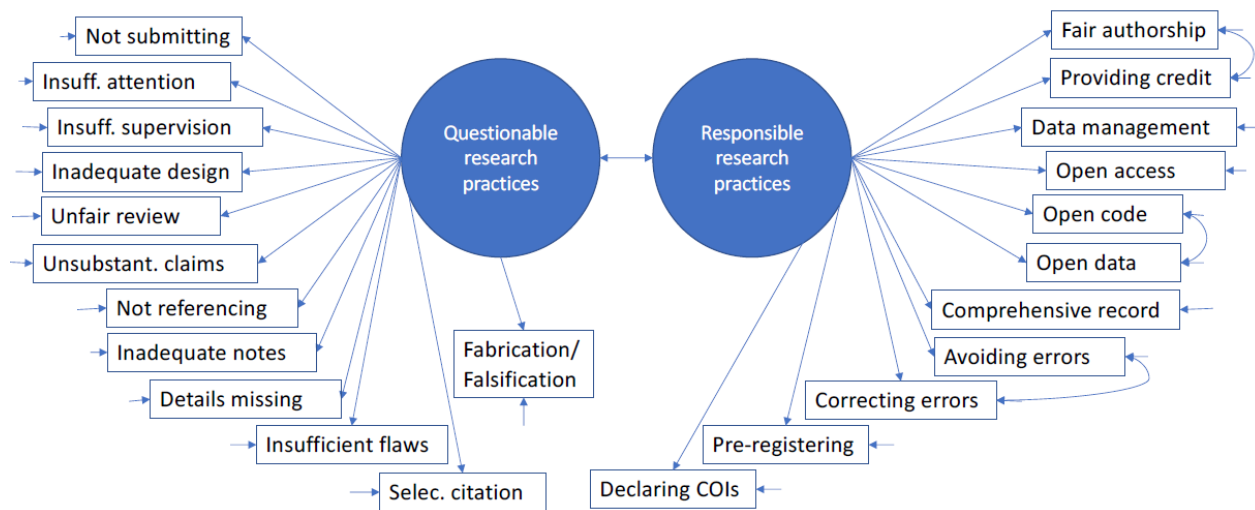
Dealing with non-applicable research practices in the MIMIC models

Recall that in our linear regressions, we replaced the answer of 1 (meaning never) for research practice items that respondents indicated not to be applicable to their research. This replacement was based on the notion that the given research practice did indeed never occur albeit for reasons of non-applicability to the research of the respondent. In the current SEM analyses, we can still meaningfully compare the groups that differ in the frequency of non-applicable answers to QRP and RRP items by freeing the relevant parameters between the groups. Inspired by the approach presented by Widaman and colleagues for completely missing indicators in some of the groups (Widaman, Grimm, Early, Robins, & Conger, 2013), we will leave out from the MIMIC model in groups those QRP and RRP items that show a percentage of Non-Applicable (NA) answers exceeding 20% that given group. In multi-group SEM we will freely estimate (without invariance constraints) the relevant measurement parameters of these items with a high prevalence of NAs to meaningfully compare groups based on the items. In this way, these items are assumed not to be invariant between the groups because of the common answer of NA in one or more groups.

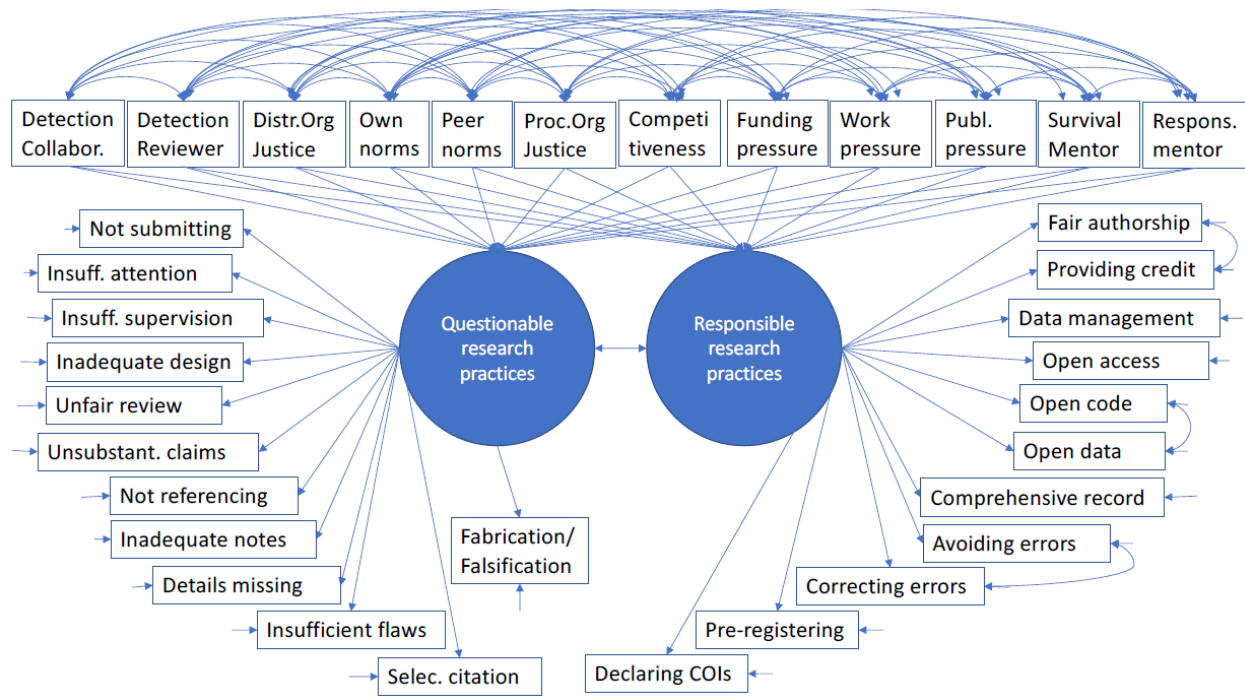
Model structures visualized



Checking a one integrity dimension solution



Checking a multi-integrity dimension solution



Bringing in the explanatory variables into the model

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